

Sam Woods – Engineering Physics

swoods07@student.ubc.ca | [LinkedIn](#) | [Portfolio](#) | (226) 501-4712

PROFILE

Second-year Engineering Physics student with hands-on experience across mechanical design, embedded electronics, and manufacturing through UBC Sailbot and personal robotics projects. Demonstrated ability to quickly pick up new tools and skills and translate technical analysis into documentation and presentations for a cross-disciplinary team. Particularly interested in problems at the intersection of mechanical systems and control, spanning marine, robotics, and aerospace applications.

TECHNICAL SKILLS

Tools/Software: SolidWorks, KiCad, Git, FEA, Arduino, Oscilloscopes

Languages: C, MATLAB, Java, Python

EXPERIENCE

Mechanical Sub-team Member

Sept 2025 – Present

UBC Sailbot Design Team – Vancouver, BC

- Co-designed a redesigned solar panel mounting system for Polaris in a 2-person team, cutting fastener count by 50% while resolving a deck clearance issue.
- Manufactured and assembled the mounting solution, using structural epoxy to bond aluminum mounting rails to the carbon fibre deck.
- Performed SolidWorks FEA on the panel mounting assembly under a 5kN worst-case wave loading scenario to validate hand-calculated fastener and mounting rail loads before manufacturing.
- Identified a galvanic corrosion risk after receiving fasteners in an incompatible aluminum alloy and implemented a mitigation strategy using PTFE thread tape to protect the mounting hardware.
- Presented design rationale and analysis to a multidisciplinary engineering team across a series of formal design reviews, incorporating feedback into iterative revisions.
- Authored internal assembly documentation for the hull, keel, and solar panel systems, enabling smaller test crews and more frequent on-water testing ahead of the vessel's first multi-day launch.
- Researching hydrofoil geometry trade-offs (T-foils vs. surface-piercing) and lift-control actuation methods for a new hydrofoiling catamaran targeting an autonomous speed record.

PROJECTS

5-DOF Robotic Arm

In progress

Personal Project

- Designed and manufactured 3 revisions of a custom ESP32 servo breakout board in KiCad to route signal and power for 6 MG996R servos, consolidating circuitry, leading to a smaller electrical footprint.
- Modeled and prototyped the mechanical structure in SolidWorks with 3D-printed components; currently developing full system integration and inverse kinematics-based control code in C++ and Python.

Robotic Claw Competition

Class Project

- Collaborated in a 6-person team to design, build, and program a servo-actuated robotic claw, engineering a joystick-probe ground detection sensor that outperformed sonar alternatives; placed 2nd out of 12.
- Produced SolidWorks CAD models and engineering drawings for the claw assembly.

EDUCATION

Engineering Physics, BSc.

University of British Columbia – Vancouver, BC

- Relevant Coursework: Linear Algebra, Calculus 1 & 2, Programming in C/C++, Electromagnetism, Thermodynamics.